

ELEC/COEN 390

Engineering team design project

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Outline

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- ▶ Understanding Language Models
- ▶ Existing Models
- ▶ Limitations and Concerns
- ▶ REST API
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Introduction to Language Models

- ▶ Language models enable machines to understand and **generate human-like text**.
- ▶ They use **neural networks** to predict the most likely next word or phrase in a given context.
- ▶ LLMs (Large Language Models) have been trained on a vast amount of text data.
- ▶ Language models like ChatGPT have applications in chatbots, virtual assistants, content creation, and language translation.

Neural Networks

[HTTPS://WWW.TIBCO.COM/REFEREN
CE-CENTER/WHAT-IS-A-NEURAL-
NETWORK](https://www.tibco.com/reference-center/what-is-a-neural-network)

Understanding Language Models

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- LLMs operate at the token level, where each token represents a discrete unit of text.
 - <https://platform.openai.com/tokenizer>
- Input text is divided into tokens, allowing the model to process and understand the context.
 - E.g.: “I love cats” becomes [“I”, “love”, “cats”]
- LLMs have a maximum token limit, such as 4096 tokens for ChatGPT.
 - Long input texts may require truncation or processing in parts.
- Token limitations can affect the model's ability to understand long contexts.

Existing Models

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- ▶ OpenAI / Microsoft: ChatGPT and related (GPT-3, GPT-3.5, GPT-4.0)
- ▶ Google (Alphabet): Gemini (LaMDA / BARD), PaLM
- ▶ Deepmind (Alphabet): Gopher, Chinchilla, Sparrow
- ▶ Meta (Facebook): LLaMA / Alpaca
- ▶ Open Source: LLaMA / Alpaca, Gemma
 - ▶ Can be run on a local machine, i.e., your home computer!
- ▶ More from other countries and companies (Mistral, Falcon, Koala, etc.)

Limitations and Concerns

- ▶ **LLMs are not sentient.** We have not created true AI (or AGI).
- ▶ **LLMs are fallible.** They can answer logical questions incorrectly, give irrational answers, and hallucinate facts and events.
- ▶ **LLMs can be manipulated.** With right inputs, an LLM can be made to generate harmful, incorrect, or contradictory information.
- ▶ **LLMs can be biased.** A model is only as good as the data it's trained on.
- ▶ **LLMs lack common sense.** A language model has never driven a car or flown a plane. It has never watched a movie or read a book. It has no memories or experiences of its own.

What is a REST API?



REST is a set of architectural constraints, not a protocol or a standard



API developers can implement REST in a variety of ways



This information, or representation, is delivered in one of several formats via HTTP:

JSON (Javascript Object Notation)

HTML

plain text



JSON is the most generally popular file format to use because, despite its name, it's readable by both humans and machines.

RESTful Criteria

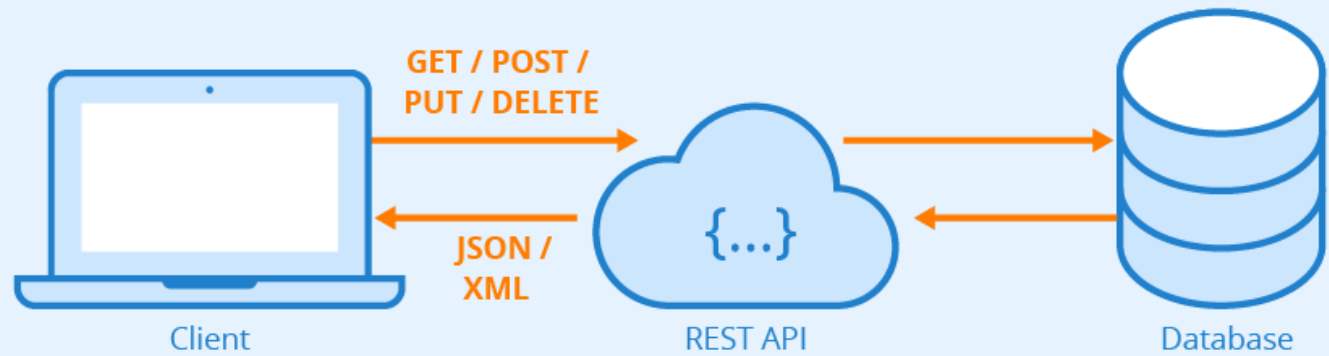
- ▶ A client-server architecture made up of clients, servers, and resources, with requests managed through HTTP.
- ▶ **Stateless** client-server communication, meaning no client information is stored between get requests and each request is separate and unconnected.
- ▶ Cacheable data that streamlines client-server interactions.
- ▶ A uniform interface between components so that information is transferred in a standard form.

HTTP Verbs

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- ▶ GET: Retrieves a representation of a resource.
- ▶ POST: Creates a new resource.
- ▶ PUT: Updates an existing resource.
- ▶ DELETE: Deletes a resource.

RESTful flow diagram



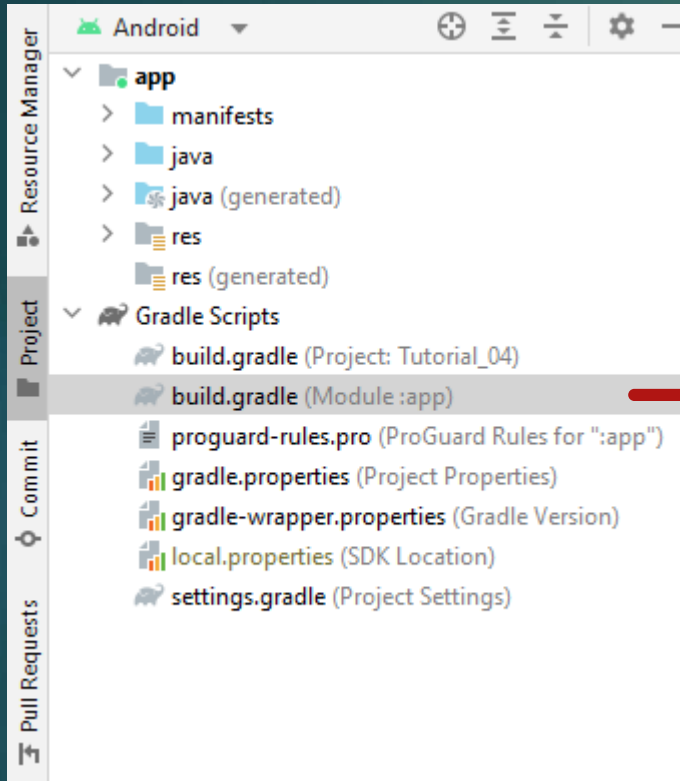
RESTful technologies

Retrofit Behind The Scenes



Step 1: Add Retrofit to your project

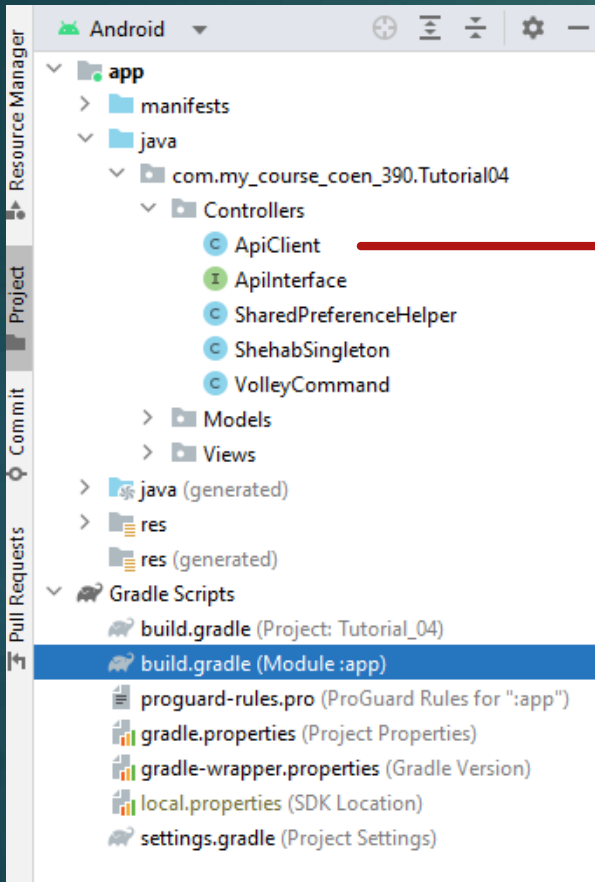
13



```
30 dependencies {
31
32     implementation 'androidx.appcompat:appcompat:1.3.1'
33     implementation 'com.google.android.material:material:1.4.0'
34     implementation 'androidx.constraintlayout:constraintlayout:2.1.1'
35     implementation 'com.squareup.retrofit2:retrofit:2.9.0'
36     implementation 'com.squareup.retrofit2:converter-gson:2.3.0'
37     implementation 'com.android.volley:volley:1.1.0'
38     testImplementation 'junit:junit:4.+'
39     androidTestImplementation 'androidx.test.ext:junit:1.1.3'
40     androidTestImplementation 'androidx.test.espresso:espresso-core:3.4.0'
41 }
```

Step 2: Define an API Client

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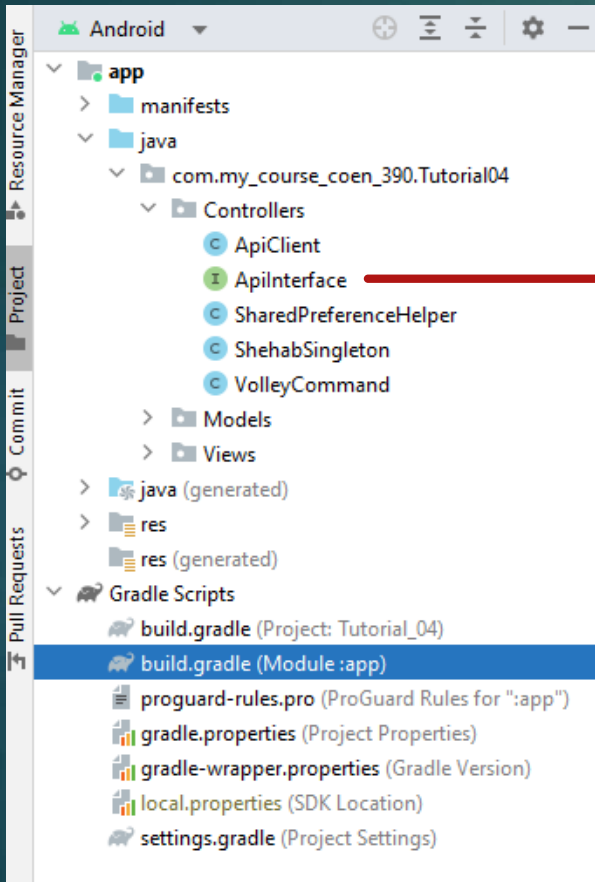
```
import retrofit2.Retrofit;
import retrofit2.converter.gson.GsonConverterFactory;

public class ApiClient {
    1 usage
    public static String BASE_URL = "https://api.openweathermap.org/";
    3 usages
    public static Retrofit retrofit=null;

    1 usage
    public static Retrofit getRetrofit()
    {
        if(retrofit == null)
        {
            retrofit = new Retrofit.Builder()
                .baseUrl(BASE_URL)
                .addConverterFactory(GsonConverterFactory.create())
                .build();
        }
        return retrofit;
    }
}
```

Step 3: Define an API interface

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```
import retrofit2.Call;
import retrofit2.http.GET;
import retrofit2.http.Query;

// Convert JSON to Java
// https://www.jsonschema2pojo.org/

public interface ApiInterface {
    End point
    @GET("data/2.5/weather")
    Call<Example> getWeather(@Query("q")String q, @Query("appid")String appid, @Query("units") String units);
}
```

Step 4: Execute request

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```
apiInterface = ApiClient.getRetrofit().create(ApiInterface.class);
Call<Example> call = apiInterface.getWeather(q: "Montreal", appKey, units: "metric");

call.enqueue(new Callback<Example>() {

    @Override
    public void onResponse(Call<Example> call, Response<Example> response) {
        Log.w( tag: "MSG_Inbox", response.message());
        if(response.isSuccessful())
        {
            Example data = response.body();

            temp.setText("Temperature " + data.getMain().getTemp());
            pressure.setText("Pressure " + data.getMain().getPressure());
            humidity.setText("Humidity " + data.getMain().getHumidity());
        }
    }

    @Override
    public void onFailure(Call<Example> call, Throwable t) {
        Toast.makeText(getApplicationContext(),
            t.getMessage(),
            Toast.LENGTH_LONG).show();
    }
});
```

step 5: json serialization and deserialization

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```
import com.google.gson.annotations.SerializedName;

public class Example {
    2 usages
    @SerializedName("main")
    Main main;

    public Main getMain() { return main; }

    public void setMain(Main main) { this.main = main; }
}
```

```
import com.google.gson.annotations.Expose;
import com.google.gson.annotations.SerializedName;

public class Main {
    2 usages
    @SerializedName("temp")
    @Expose
    private Double temp;
    2 usages
    @SerializedName("pressure")
    @Expose
    private Integer pressure;
    2 usages
    @SerializedName("humidity")
    @Expose
    private Integer humidity;
    2 usages
    @SerializedName("temp_min")
    @Expose
    private Double tempMin;
    2 usages
    @SerializedName("temp_max")
    @Expose
    private Double tempMax;
```

JSON response

```
"main": {
  "temp": 298.48,
  "feels_like": 298.74,
  "temp_min": 297.56,
  "temp_max": 300.05,
  "pressure": 1015,
  "humidity": 64,
  "sea_level": 1015,
  "grnd_level": 933
},
```

DEMO

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- ▶ Using ChatGPT and okhttp

System Prompts

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- System prompts are initial text inputs provided to a language model to guide its responses.
- They set the context and tone for the conversation, influencing the behavior of the model.
 - They can be used to specify the role or persona of the model, establish a specific topic, or **provide instructions**.